# NorthRiver Business Case — Maintenance Knowledge & Records Assistant

Oct 25, 2026 · @Ojowu Omakwu

## Summary and recommendation

**Recommendation:** approve a bounded, read-only pilot of a governed maintenance knowledge and records assistant, together with rationalization of the station maintenance document library. The use case ranked first of six on NorthRiver's Use-Case Prioritisation Scorecard (16.7 of 25, 5.9 points ahead of the next candidate) and is the only one ready to pilot.

- **What it does:** answers maintenance staff's questions about standards, procedures and bulletins with cited sources, checks the answer against the asset's record in the EAM system, and declines when the evidence doesn't support an answer.
- **Cost (illustrative):** about CAD 156,000 of one-time effort for a 12-week pilot in one operating area, plus about CAD 600–1,100 a month to run.
- **Value (target, not evidence):** about CAD 184,000 a year of staff capacity for every hour a week saved across 40 planners and engineers. The one-time cost is covered in a year at 0.85 hours a week per person.
- **Risk:** Medium tier under NorthRiver's rules. Answers inform work on energized equipment, but the assistant is advisory and handles little personal data.

**Decision requested:** approve pilot preparation, the Medium-tier governance route, and the pilot funding envelope (see Decisions requested and next steps).

*All figures describe a fictional organization and are illustrative. Benefit figures are targets to be tested, not results.*

## Problem and strategic fit

Maintenance staff can't quickly find the correct, current standard for a specific asset and check it against that asset's record in the EAM system. Planners and asset engineers spend an estimated 4–6 hours a week each doing this (illustrative planning assumption).

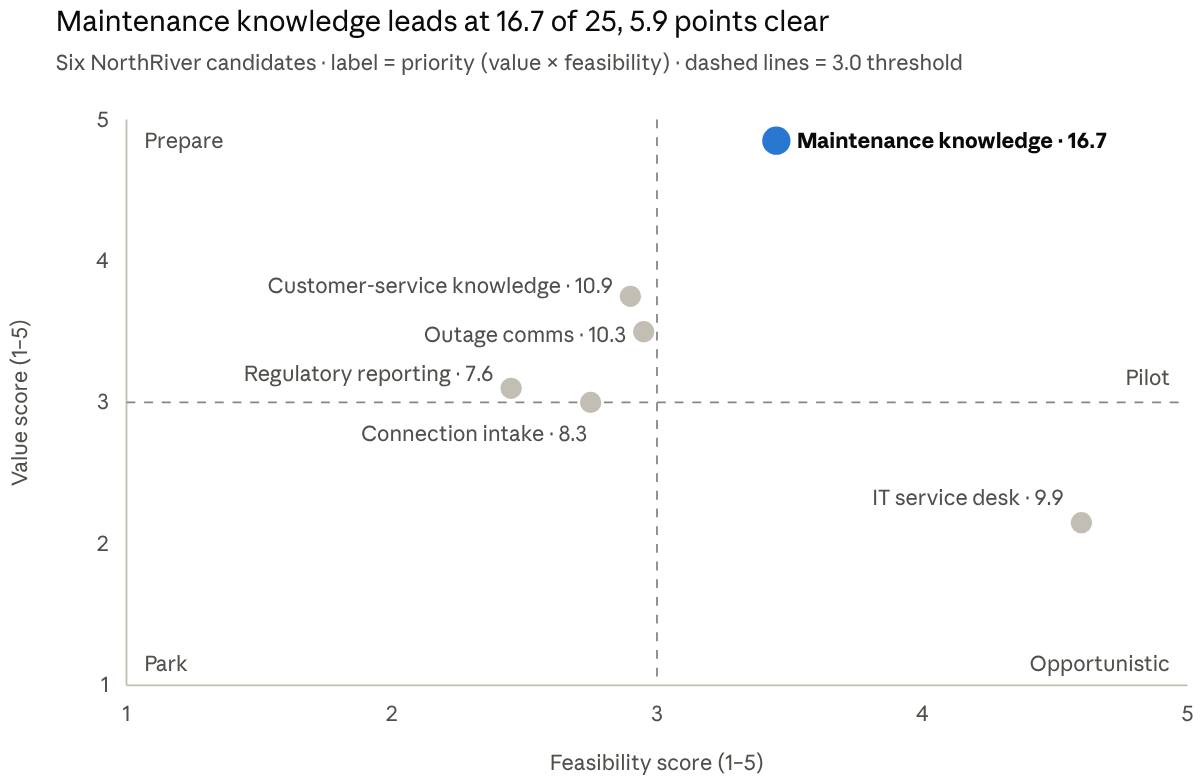
Three things make it worse at NorthRiver:

- **Two utilities' documents.** Since the 2019 merger with the former Lakeshore Hydro, both utilities' standards remain on file, with overlapping titles and different intervals.
- **Exceptions live outside the standards they change.** An approved exception or bulletin for a specific asset class is easy to miss.
- **Expertise is retiring.** The people who know which rule applies to which older asset are the people closest to retirement.

The results are slow planning, inconsistent decisions between operating areas, compliance evidence rebuilt after the fact, and dependence on a shrinking group of experts. Full framing: `NR_Flagship_Problem_Statement.md`.

**Strategic fit.** This is the first use case through NorthRiver's AI adoption value stream. Its selection (this business case and the Scorecard) is stage 2, Assess, Prioritize and Decide. The retrieval-plus-record pattern it proves is reusable by later use cases, which is why V3 (reuse value) scored 5.

## Selection evidence




NR Use-Case Prioritisation Scorecard v1.0, Scoring sheet · illustrative, self-assigned scores

Maintenance knowledge is the only candidate in the Pilot quadrant. Criteria, weights, the 3.0 threshold and the risk-tier rules were fixed before any candidate was scored, and the reason for every score is on the Scorecard's Rationale sheet.

- **The result holds under reweighting.** It stays first with equal criterion weights. The IT service desk overtakes it only if feasibility carries more than 70% of the decision, which is the "start with the easiest" argument made explicit.
- **Four of six candidates need preparation first.** Customer-service knowledge ranks second but is High tier because customer account data runs through it; privacy controls must exist before it starts.
- **The winner's weak point is feasibility (3.45).** Messy two-utility documents and the EAM connection are why Option B and the pilot bounds are part of the recommendation.

## Options considered

Option D with Option B is recommended. B is needed whatever else happens; D is the only option that links documents to EAM records with controls fit for maintenance-critical answers.

| Option | Links documents to EAM records | Current vs. superseded control | Evaluation gates for critical answers | Time to value (weeks from approval, illustrative) | Assessment |
|---|---|---|---|---|---|
| A. Do nothing | No | No | No | — | Worsens with each retirement. Rejected |
| B. Rationalize the whole document library + better search | No | Yes, once cleaned | No | 26–52 weeks for the whole library | Needed regardless. Only its pilot scope is a dependency for D. Partial |
| C. Structured expert knowledge capture | No | Partly | No | 26+ weeks; continuous | Complementary; captured knowledge still needs to be findable. Partial |
| D. Governed retrieval assistant + read-only EAM lookup, alone | Yes | Yes (authority metadata) | Yes (EG-01 to EG-16) | Not run alone: needs clean pilot documents | Would put a messy corpus in front of the evaluation gates |

| Option | Links documents to EAM records | Current vs. superseded control | Evaluation gates for critical answers | Time to value (weeks from approval, illustrative) | Assessment |
|---|---|---|---|---|---|
| **D + B, pilot scope (recommended)** | **Yes** | **Yes** | **Yes** | **About 12 weeks to first pilot use; about 24 weeks (6 months) to measured results** | **Recommended.** A two-week discovery fixes the document schema first; then clean-up and build run in parallel |
| E. Microsoft 365 Copilot over the document repository | No | Limited | Limited | 4–6 weeks to first use | Fastest start, but no governed EAM lookup or authority control. Revisit if those become configurable |
| F. AI that plans or schedules maintenance | Yes | — | — | 52+ weeks | Unacceptable on safety-critical assets; conflicts with human accountability. Rejected |

**Option E is the comparison that matters most.** The Workflow Transformation Artifact should test it with evidence (a small spike on the same questions) rather than settle it by argument. If Copilot can be configured to meet the Tier A evaluation gates, the build case weakens and that is recorded.

## Costs

A 12-week pilot costs about CAD 156,000 of one-time effort, almost all internal staff time, plus CAD 600–1,100 a month in cloud services. Figures are illustrative: effort at an assumed loaded internal rate of CAD 100 an hour, cloud at approximate list prices to be confirmed at pilot time.

### One-time pilot effort

| Work | Person-weeks | Cost (CAD) |
|---|---|---|
| Discovery: document inventory, sample profiling, metadata schema, EAM data profiling (2 weeks) | 3 | 12,000 |
| Document rationalization for the pilot scope (Option B) | 12 | 48,000 |
| EAM connector, assistant build, hosting setup | 10 | 40,000 |
| Evaluation and governance (test set, EG gates, AIA-lite, control record) | 6 | 24,000 |
| Subject-matter expert time (writing and judging test questions) | 4 | 16,000 |
| Change, training and the user test | 4 | 16,000 |
| **Total** | **39** | **156,000** |

**Monthly run cost during the pilot**

| Item | CAD per month |
|---|---|
| Azure AI Search (Basic tier, semantic ranker available) | ~100 |
| Azure OpenAI (embeddings and generation at pilot volumes) | 300–800 |
| Hosting, logging, key management | 200 |
| **Total** | **600–1,100** |

**Beyond the pilot (order of magnitude).** Enterprise rollout would add a larger search tier, higher model usage and about 0.3 FTE of ongoing support and content governance. That is sized in the production-readiness section of the reference architecture, not committed here.

**Not included:** licences NorthRiver already holds (Microsoft 365, Entra ID), and the separate Copilot Studio Use-Case Assessment Agent.

# Benefits and how they will be measured

The case rests on three kinds of value: staff capacity, consistency and risk reduction, and knowledge retention. Only capacity is expressed in dollars, and it is released time, not cash savings. Every figure below is a target until the pilot measures it.

**Capacity (illustrative).** 40 planners and asset engineers × 46 working weeks × CAD 100 an hour = about CAD 184,000 a year for each hour a week saved per person.

- Break-even: the CAD 156,000 one-time cost is covered in one year at **0.85 hours a week** per person.
- Target: halve the estimated 4–6 hours a week, which would release CAD 368,000–552,000 a year of capacity across the 40.

| Benefit | Measure | Baseline method (before the pilot) | Pilot target |
|---|---|---|---|
| Faster access to the applicable standard | Minutes to find and confirm the standard for a set task | Timed task test: 10 standard questions, pilot group, without the assistant | 40% faster with it |
| Consistent answers across operating areas | Same question, same cited answer | Same 10 questions asked by staff in two areas | Repeatability EG-11 ⩾ 90%; 100% on critical facts |
| Trustworthy answers | Unsupported or wrongly cited claims | n/a (new capability) | Tier A gates pass (EG-02 to EG-04, EG-08) |
| Current guidance wins | Superseded content presented as current | Audit sample of recent work packages | Zero (EG-05) |
| Compliance evidence as a by-product | Time to reconstruct why a decision was made | One past audit query, timed | Answer from the decision log in minutes |
| Adoption | Weekly active users in the pilot group; reviewer overrides | n/a | ⩾ 70% active by week 8; override rate tracked (EG-13) |

**Not claimed:** onboarding time (18–24 months today) and retention of retiring experts' knowledge. Both are real but need longer than a pilot to show; they are tracked in the benefits model, not counted here.

**Evidence rule.** Synthetic prototype results show whether the design works. They are never reported as realized NorthRiver benefits.

## Risks and governance

The use case is **Medium risk tier**: consequence of a wrong answer 4 of 5, personal-information exposure 2, decision autonomy 2. Under NorthRiver's tier rules that means approval by the AI governance lead and the business owner, the full Tier A and B evaluation gates before the pilot, an AIA-lite, and a semi-annual review.

| Risk | Effect if it happens | Control | Evidence |
|---|---|---|---|
| Unsupported answer on a safety-critical procedure | Staff act on wrong guidance near energized equipment | Grounded-or-refuse; mandatory citations; advisory only | EG-03, EG-04 (Tier A); NR-AIGOV-002 |
| Superseded or unapproved document presented as current | Wrong interval or method applied | Authority metadata; conflict disclosure | EG-05, EG-09 (Tier A) |
| Customer personal information reaches an unentitled role | Privacy breach | Field-level filtering in the EAM connector; never logged | EG-06 (Tier A): any failure is NO-GO |
| EAM or search unavailable | A plausible but invented status | Controlled failure: answer the standards part, say the record is unavailable | EG-12 (Tier A) |
| Over-reliance on the assistant | Staff stop checking | Citations shown with every answer; training; override tracking | EG-13; adoption plan |

| Risk | Effect if it happens | Control | Evidence |
|---|---|---|---|
| Low adoption or staff distrust | Benefits don't materialize | Field staff in the user test; union engaged early; tool supports, doesn't monitor | Adoption measures |
| Document rationalization slips | Pilot starts on a messy corpus | Discovery gate fixes the schema first; pilot scope limited to cleaned documents | Corpus version recorded per test |
| Cloud provisioning or cost overrun | Delay or budget pressure | Provider abstraction; usage limits; cost per query tracked | EG-15 |

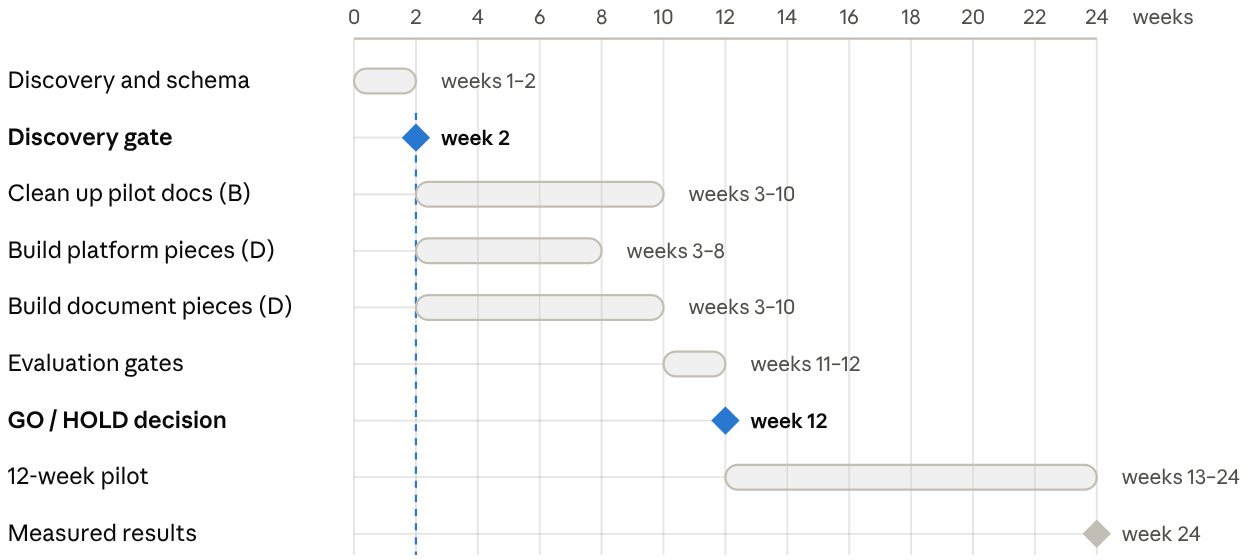
**Decision logic.** Evaluation results become GO, GO WITH CONDITIONS, HOLD or NO-GO under the Evaluation-to-Governance Mapping. No average score can override a Tier A failure.

## Pilot scope, success and stop criteria

A 12-week, read-only pilot with planners in one operating area and one station maintenance crew, after the evaluation gates pass on the pilot corpus.

**Discovery gate at week 2, first pilot use at week 12**

Illustrative · weeks from approval · document-dependent build starts only after the schema is agreed




Illustrative pilot plan · weeks from approval · Option D with Option B (pilot scope)

A two-week discovery settles how the pilot documents are versioned, approved and formatted, and how reliable the EAM data is, before any document-dependent build starts. After that gate, clean-up and build run side by side; the evaluation set and retrieval tuning use cleaned documents only.

**In scope:** maintenance standards, procedures, bulletins and practice notes for municipal stations and their primary equipment; read-only lookup of asset and work-order records.

**Out of scope:** any write to the EAM system; scheduling or planning decisions; real-time operational data; customer-facing use.

**Success at week 12 (all must hold):**

- Every Tier A gate passes on the final evaluation run (GO, or GO WITH CONDITIONS with named owners).
- Timed task test at least 40% faster than baseline.
- At least 70% of the pilot group active weekly by week 8.
- No unresolved material limitation left out of the limitations register.

**Stop (HOLD or NO-GO for the affected scope):**

- Any customer personal information shown to an unentitled role (EG-06).
- A superseded procedure presented as current (EG-05).
- A confirmed case of staff acting on an uncited or wrong answer.

- Groundedness below the agreed threshold on the evaluation set.

## Assumptions, limitations and evidence boundary

NorthRiver is fictional, so this case demonstrates method, not a real investment decision.

**Assumptions**

- 40 planners and asset engineers use maintenance standards weekly; 46 working weeks a year; CAD 100 loaded hourly rate.
- The EAM system can expose read access through an API or export.
- The pilot document set can be rationalized in about 12 person-weeks.
- NorthRiver's Microsoft environment (Entra ID, Azure, Microsoft 365) is available to host the pilot in Canadian regions.

**Limitations**

- Scorecard scores are self-assigned. In a real organization they would be set with the sponsor, domain owners, IT, privacy and risk.
- Time savings are planning assumptions until the baseline task test is run.
- Applicable privacy law and Ontario AI guidance are confirmed in the governance phase.

**Evidence boundary**

| Claim | Status |
|---|---|
| Scorecard method, scoring and sensitivity | Built (illustrative inputs) |
| Architecture, controls, evaluation gates | Designed |
| Working prototype and evaluation results | In development; synthetic data |
| Pilot costs and benefits | Proposed; illustrative |
| Realized NorthRiver benefits | Not claimed |

## Decisions requested and next steps

**Decisions requested from the sponsor (VP Asset Management & Grid Planning)**

1. Approve pilot preparation for Option D with Option B.

2. Confirm the Medium-tier governance route: AI governance lead and business owner approve; full Tier A and B gates before the pilot.
3. Approve the illustrative funding envelope: CAD 156,000 one-time effort plus up to CAD 1,100 a month during the pilot.
4. Name a document owner for the pilot document set.

**Next steps**

| Step | Output | When (weeks from approval) |
|---|---|---|
| Discovery: inventory and profile the pilot documents; profile EAM data quality | Agreed metadata schema, authority rules and format handling; data-quality findings | Weeks 1–2 |
| Workflow Transformation Artifact, including the Option E spike | Current and target workflow, baselines, evidence on Copilot vs. build | Weeks 1–2 |
| **Discovery gate** | Schema agreed; Option D confirmed over E, or the plan changes | **End of week 2** |
| Build platform pieces: model and search adapters, EAM connector, decision log, evaluation harness | Domain-independent components | Weeks 3–8 |
| Rationalize the pilot document set (Option B, pilot scope) | Current, versioned corpus with authority labels | Weeks 3–10 |
| Build document pieces: corpus adapter, chunking, retrieval tuning on cleaned samples | Assistant working on the pilot scope | Weeks 3–10 |
| Baseline task test | Timed baseline for the 10 standard questions | Weeks 8–11 |
| Control record NR-AIGOV-002 and AIA-lite | Control evidence for the gate | Before week 12 |
| Evaluation on the cleaned corpus | GO / HOLD decision, failures kept | Weeks 11–12 |

| Step | Output | When (weeks from approval) |
|---|---|---|
| Pilot, user test and measured results | Adoption and time-to-answer results against targets | Weeks 13–24 |